30 MAY 1979
SUPERSEDING
MIL-C-83346 (USAF)
20 December 1971

MILITARY SPECIFICATION

CHAFF, COUNTERMEASURES RR-113/AL

"This specification is approved for use by the Air Force Logistics Command, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense." *

1. SCOPE

1.1 Scope. This specification covers one type of radar reflector, designated Chaff, Countermeasure RR-113/AL.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein. *

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-B-131	Barrier Material: Water Vaporproof, Flexible
MIL-R-5253	Reflectors, Radar, General Specification for *

STANDARDS

Military

MIL-STD-831	Test Reports, Preparation of

DRAWINGS

U.S. Air Force

50B12985	Case - Strip Reflector, Inner
59D4414	Sleeve - Outer
60C2522	Box, Folding, Shipping and Dispensing, Air Force
60B2523	Reflector - Strip
63C3129	Support Sleeve - Countermeasures Chaff

(Copies of specifications, standards, drawings, and publications required by * contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 5865*

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply. *

American Society for Testing Materials

DD689-62 Standard Method of Test for Internal Tearing Resistance of Paper*
D774-67 Standard Method of Test for Bursting Strength of Paper *

Applications for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia, PA 19103. *

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 General specification. The requirements of MIL-R-5253 apply as requirements of this specification with the exceptions and additions, identified by similarity of paragraph headings, called out herein. When the two documents conflict, this specification shall govern.

3.2.1 Foil finish, nominal thickness, and temper. The foil finish shall be plain. The nominal thickness for cut 1 shall be 0.0018 inch, and the temper shall be H-18 of 3005 alloy. The nominal thickness for cuts 2 through 4 shall be 0.00045 inch, and the temper shall be H-18 of 1145 alloy.

3.2.2 Coatings. The dipoles shall be slip coated and stripe coated as specified herein. Slip coating shall be applied over stripe coating.

3.2.2.1 Slip coating. Slip coating shall be applied to the chaff material to: (1) prevent dipole end welding and metal smear during the cutting operation, (2) prevent cohesion between dipoles which may be caused by corrosive conditions, and (3) increase dispersion by reducing dipole-to-dipole friction. *

3.2.2.1.1 Composition. The slip coating shall be composed of the following ingredients, or their equivalent as approved by the procuring activity:

a. Eighteen fluid ounces of Silicone SM-2016, General Electric Company or Silicone Syl-Off 22, Dow Corning Corporation, or equal, 40 percent aqueous emulsion. *

b. Eighteen fluid ounces of Silicone SM-2010, General Electric Company, or* Silicone DC-24, Dow Corning Corporation, or equal, 35 percent aqueous emulsion.

c. Three and one-half ounces of aluminum acetate, 20 percent aqueous solution, sulfate free. *

d. Seven gallons of filtered water.

NOTES: (1) The ingredients shall be mixed at the time of use; the reaction

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shall take place on the foil at the time of application.

(2) The coating formula may necessitate modification to be adaptable to different manufacturer's application equipment.

3.2.2.1.2 Application. The slip coating composition shall be applied to both surfaces of the dipoles.

3.2.2.2 Stripe coating. The stripe coating referenced in MIL-R-5253 shall be referred to herein as stripe coating.

3.2.2.2.1 Composition. Composition of the stripe coating shall conform to MIL-R-5253. The stripe coating weight shall be 0.0225 pound (±20 percent) per 1000 square inches of area covered.

3.2.2.2.2. Application. The stripe coating shall be continuously and uniformly applied on one surface of the dipoles. The stripe coating shall start within 10 percent of one end of the dipole and the coated length shall be as follows (±10 percent):

Cut	Dipole length (inches)	Coated length (inches)
1	12.92	None
2	4.42	None
3	2.00	0.084
4	0.98	0.140

3.3 Design and construction. The RR-113/AL chaff unit shall be of airborne passive electronic countermeasures material. The chaff unit shall be so packaged that it can be dispensed from the AN/ALE-24 and AN/ALE-27 * dispensers.

3.3.1 Recycled, virgin and reclaimed materials. The use of reclaimed material shall be encouraged to the maximum extent possible.

3.3.2 Unit design data. The chaff unit shall be composed of aluminum foil dipoles packaged in inner cases. Cut 1 shall consist of flat dipoles. Cuts 2 through 4 shall be V-bend dipoles.

3.3.2.1 Dipole quantities. The dipoles shall be cut in 4 lengths in nominal quantities as follows:

Cut	Dipole Length (Inches)	Dipole Quantity Per Inner Case	No. of Inner Cases Per Unit	Total Dipoles Per Unit
1	12.92	1,500	1	1,500
2	4.42	18,000	1	18,000
3	2.00	80,000	1	80,000
4	0.98	80,000	2	160,000

3.3.2.1.1 Dipole quantities per inner case may be adjusted to meet the requirements of 3.3.4.3.1. The quantities shall be packed in the inner cases to preclude dipoles sticking together.

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3.3.3 Dipole dimensions. The aluminum foil dipoles shall be constructed in accordance with Drawing 60B2523. The flat dipoles shall be constructed without the V-bend. The dipole dimensions shall be as follows (in inches):

Cut	Dipole Length	Dipole Width	Dipole Thickness
1	12.92	0.016	0.0018
2	4.42	0.016	0.00045
3	2.00	0.008	0.00045
4	0.98	0.008	0.00045

3.3.4 Packaging

3.3.4.1 Inner cases. The chaff unit shall contain five inner cases. The dimensions of the inner case shall be determined by the supplier consistent with the requirements of 3.3.4.3.1.

3.3.4.1.1 The inner cases as specified in 3.3.4.1 shall be constructed in accordance with Drawing 50B12985 and shall contain the quantity of dipoles as specified in 3.3.2.1. The inner cases shall be constructed from 30- to 60-pound Natural Kraft, Number 1 Double Glaze (0.005-inch nominal thickness). The inner cases shall be perforated and each face shall form 90-degree angles to each other.

3.3.4.1.2 Arrangement of inner cases. The inner cases shall be arranged in three layers in the assembled RR-113/AL chaff unit. The top layer shall contain cut 1, the middle layer shall contain cut 2, and the bottom layer shall contain cuts 3 and 4.

3.3.4.2 Support sleeve. The material for the chaff unit support sleeve shall be high flex, medium impact polystyrene and, as part of the assembled unit, shall withstand the environmental conditions specified under 3.4.1. The support sleeve shall conform to Drawing 63C3129 and shall be constructed to preclude warpage and breakage.

3.3.4.3 Outer case (sleeve)

3.3.4.3.1 The outer case shall conform in shape, markings, material and assembled size to Drawing 59D4414. The detailed dimensions of Drawing 59D4414 shall be adjusted to obtain a completely tight assembled unit. The V-bend dipole quantity specified in 3.3.2.1 may be adjusted to obtain a tightly assembled unit.

3.3.4.3.1.1 When assembled, the outer case shall show no light between the hourglass flap and folded flaps M and N.

3.3.4.3.1.2 When a 20-pound force is applied to a column of 30 assembled units, representing loading in the AN/ALE-24 dispenser magazine, the outer case width and thickness shall conform to the assembled dimensions on Drawing 59D4414.

3.3.4.3.2 The outer case color shall conform to the requirements of Drawing 59D4414.

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3.3.4.3.3 Outer case closure. The outer case shall be held closed in accordance with Drawing 59D4414.

3.3.4.3.3.1 Sealing. The sealing compound shall be an aqueous type adhesive polyvinyl acetate. When applied to the outer case, the sealing compound shall be impervious to the environmental conditions specified under 3.4.1.

3.3.4.3.4 Material strength. The tear strength of the outer case shall be 424 grams against the grain and 192 to 224 grams with the grain at -65°F. The burst strength of the outer case shall be 91 to 105 pounds per square inch at -65°F.

3.3.4.3.5 The outer case shall be so constructed that, when placed in columns of 30 assembled units and subject to the environmental conditions specified herein, no outer case shall discolor, delaminate, adhere to another, or corrode.

3.3.4.4 Arrangement of assembled chaff unit. The Chaff, Countermeasures RR-113/AL shall be assembled in accordance with the requirements herein.

3.4 Performance. The Chaff, Countermeasures RR-113/AL shall produce confusing signals received in radar systems operating in the frequency range of 425 to 5900 MHz. The chaff unit shall simulate an airborne vehicle whose radar amplitude is 100 square meters.

3.4.1 Environmental conditions.

3.4.1.1 The RR-113/AL chaff shall withstand without degradation to equipment performance the following environmental conditions:

a. Temperature: -85°F to +185°F, including room ambient temperatures.

b. Vibration: Frequencies varying from 5 to 30 Hz at amplitudes varying 0.02 to 0.20 inch double amplitude to those corresponding to acceleration levels to 0.5g.

c. Humidity: Room ambient conditions up to 95 percent relative humidity, including condensation.

3.4.1.2 When loaded as follows, the chaff units shall be ejectable after being subjected to the environmental conditions of 3.4.1.1:

a. Thirty (30) assembled RR-113/AL chaffs loaded in an AN/ALE-24 dispenser magazine under a feeder force of 20 pounds applied at the aft end of the magazine and orientated in the manner as installed in the aircraft.

b. Thirty-eight (38) assembled RR-113/AL chaffs loaded in an AN/ALE-27 dispenser magazine under a feeder force of 5 pounds and orientated in the manner as installed in the aircraft.

3.5 Weight. The weight of the RR-113/AL chaff shall not exceed 8 ounces.

3.6 Identification of product. For identification purposes, the outer sleeve of the RR-113/AL chaff shall be permanently stamped with the following information:

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a. Type designation: RR-113/AL characters shall be 3/8 inch bold face type in black waterproof ink.

b. Date of manufacture: A 6-digit number designating the day first, the month second, and the year last; e.g., 1 December 1971 would be shown as 011271. Characters shall be 3/16 inch bold face type in black waterproof ink.

c. Manufacturer's code: A 3-letter code to be obtained from the procuring activity. Characters shall be 3/16 inch bold face type in black waterproof ink.

NOTE: The identification marking shall be stamped parallel to the longitudinal axis of the unit and on the opposite side of the hourglass face.

3.7 Workmanship and finish. Plus or minus tolerance accumulations in one direction shall not preclude the requirements of workmanship and finish for a unit free of both crowding and soft spots.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 The inspection and testing of the Chaff, Countermeasures RR-113/AL shall conform to MIL-R-5253 and the requirements specified herein. When the two documents conflict, the requirements of this specification shall govern.

4.3 Preproduction inspection.

4.3.1 Design data. The preproduction test report shall be submitted to the procuring activity in accordance with MIL-STD-831.

4.3.2 Preproduction tests.

4.3.2.1 Thermal cycling and humidity tests. Thirty assembled units loaded in an AN/ALE-24 dispenser magazine and thirty-eight assembled units loaded in an AN/ALE-27 dispenser magazine shall be subjected to the thermal cycling and humidity tests specified in MIL-R-5253. The total consecutive thermal cycles shall be 10.

4.3.2.2 Vibration-humidity test. At the conclusion of the tests specified in 4.3.2.1, the 68 assembled units (in respective magazines) shall be subjected to the vibration-humidity tests specified in 4.4.2.2.1.2.

4.3.2.2.1 After environmental testing, the assembled units shall be subjected to the requirements of 4.4.2.2.1.3 and 4.4.2.2.1.4.

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4.4 Acceptance inspection.

4.4.1 Individual tests.

4.4.1.1 Conformance to drawings. The supplier shall establish an inspection method, subject to the approval of the procuring activity, that will assure conformance of final assembly of the RR-113/AL chaff unit to the design established by the supplier to meet the requirements herein.

4.4.1.1.1 The supplier's inspection and assembly procedure shall, as a mini-* mum, ensure the following:

a. Uniformity of package density

b. Uniformity of package dimensions

c. Uniformity of package weight

d. Correct package assembly

e. Correct overall size

f. Proper positioning of the supports in each assembled package

g. Proper folding of the flaps

h. Tightness of the outer case around supports and inner cases after the chaff unit has been sealed.

4.4.2 Sampling tests.

4.4.2.1 Outer case (sleeve).

4.4.2.1.1 Water repellency tests.

4.4.2.1.1.1 Water repellency tests (general). A 4- by 4-inch sample of the outer case material shall be checked for water repellency by gradually dropping 250 cc of cold water (70°F) on a colored foil side of the sample. This procedure shall be repeated using hot water (180°F). The sample shall be visually inspected; the outer case material shall be considered unacceptable for production use if any of the following is noted:

a. Water beads do not form and roll off.

b. Foil blistering.

c. Delamination of the foil from the Kraft board.

d. Discoloration or corrosion of the foil.

4.2.2.1.1.2 Water repellency test (thermal stability). A sample of a colored outer case shall be placed in an oven at 185°F for 24 hours and allowed to cool to room temperature. The sample shall then be subjected to the cold and hot water testing procedures of 4.4.2.1.1.1. Acceptability of the outer case material shall be visually determined as specified in 4.4.2.1.1.1.

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4.4.2.1.2 Material strength tests

4.4.2.1.2.1 Tear strength. The tear strength of the outer sleeve shall be tested in accordance with ASTM D689-62. *

4.4.2.1.2.2 Burst strength. The burst strength of the outer sleeve shall be tested in accordance with ASTM D774-67. *

4.4.2.1.2.3 The tear and burst strength tests specified in 4.4.2.1.2.1 and 4.4.2.1.2.2, respectively, shall not preclude similar tests as specified in MIL-R-5253.

4.4.2.2 Assembled units.

4.4.2.2.1 Environmental tests. After each 250-carton production, the supplier shall subject 30 assembled units in an AN/ALE-24 dispenser magazine and 38 assembled units in an AN/ALE-27 dispenser magazine to tests specified in 4.4.2.2.1.1 and 4.4.2.2.1.2.

4.4.2.2.1.1 Thermal cycling and humidity tests. Thermal cycling and humidity tests shall be conducted in accordance with MIL-R-5253, except that the total consecutive cycles for the thermal cycling shall be 10. "Reflectors" called out in the tests shall be referred to herein as "assembled units."

4.4.2.2.1.2 Vibration-humidity tests. At the conclusion of the thermal cycling and humidity tests, the assembled units shall be subjected to a vibration-humidity test under the following conditions:

a. Vibration envelope.

(1) 5 to 7 Hz: 0.20 inch double amplitude

(2) 7 to 22 Hz: $\pm$0.5g

(3) 22 to 30 Hz: 0.02 inch double amplitude

b. Sweep cycle: One sweep cycle shall be from 5 to 30 Hz back to 5 Hz at the amplitudes specified in 4.4.2.2.1.2a.

c. Sweep duration: One sweep cycle shall take not less than 30 minutes. The sweep rate or the rate of frequency change shall be proportional to frequency.

d. Ambient environment.

(1) Temperature: 80°F to 110°F

(2) Humidity: 95 percent relative

The vibration test fixture and the magazines of chaff shall be enclosed in a suitable chamber and the temperature and humidity conditions specified in 4.4.2.2.1.2d shall be maintained during all sweep cycles. A total of 5 sweep cycles shall be made in each of 3 mutually perpendicular axes.

4.4.2.2.1.3 At the conclusion of the environmental testing, the assembled units shall be inspected as follows:

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a. Support sleeve.

(1) Signs of warpage or breakage.

(2) Conformance to Drawing 63C3129.

b. Outer sleeve.

(1) Signs of corrosion, adhesion, or unsealing.

(2) Conformance to Drawing 59D4414.

c. Conformance of the assembled units to the assembled dimensional tolerances specified herein.

The corresponding magazines of units shall then be ejected from the AN/ALE-24 and AN/ALE-27 dispensers, respectively, without causing malfunction to the dispensers.

4.4.2.2.1.4 Rejection and retest. If the assembled units do not meet the requirements of the environmental tests specified under 4.4.2.2.1 or show deficiencies during the inspection of 4.4.2.2.1.3, the supplier shall determine and correct the deficiencies and rerun the environmental tests prior to proceeding with further production.

4.4.3 Pre-package test. Prior to being packaged for shipment, all acceptable assembled chaff units shall be placed horizontally in columns of 30 (as installed in the AN/ALE-24 dispenser magazine). Each column shall be subjected to a 50-pound feeder force. A moving scale with a width of 3.050, +0.025, and -0.000 inch (inside width of the AN/ALE-24 dispenser magazine), shall be passed along each column of units. The scale shall move freely without making surface contact with any of the 2-7/8-inch width chaff units.

4.4.3.1 Rejection. If the moving scale makes surface contact with any chaff unit of the column, this shall be rejected for shipment.

5. PACKAGING *

5.1 Chaff, Countermeasures RR-113/AL shall be prepared for delivery in accordance with Section 5 of MIL-R-5253 and the requirements specified herein. *

5.2 Units per carton (Air Force Procurement). Unless otherwise specified, the number of tapeless units per carton shall be 120. The carton shall contain 3 layers with 40 units per layer. Each layer shall consist of 2 equal rows of 20 units. All units loaded into the inner carton shall be so placed that the hourglass face of the unit is oriented toward the front of the inside carton, with the sealed side of each unit facing the top of the carton and with the longitudinal axis of the unit parallel to the bottom of the carton.

5.3 Inside carton dimensions (Air Force). The nominal inside carton dimensions shall be in accordance with Drawing 60C2522. The carton height shall accommodate the required number of units including the separators that shall be inserted beneath the bottom layer of units, between each layer of units, and on top of the uppermost layer of units. Separators shall be of 5- to 20-point board and their dimensions shall conform to the carton length and

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width to within 1/4 inch.

5.3.1 Inner carton. The inner carton containing the individual units of chaff shall be enclosed in a close-fitting heat-sealed bag, fabricated from barrier material conforming to MIL-B-131, Class 1. The tests shall be in accordance with method IA-15 of MIL-P-116. The air volume within the barrier* material shall be reduced to a practicable minimum before sealing the barrier.

5.4 Shipment marking nomenclature. The shipment marking nomenclature shall be "Chaff, Countermeasures RR-113/AL."

6. NOTES

6.1 Intended use. Chaff, Countermeasures RR-113/AL is intended for use as a radar countermeasure for training purposes to produce confusing signals received in radars operating at 425 to 450 MHz, 1270 to 1330 MHz, 2700 to 2900 MHz, and 5400 to 5900 MHz.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Number of preproduction samples and point of inspection, and concurrent delivery of each sample and its test data. The procuring activity should be given at least ten days prior notice when the preproduction tests are to be conducted so that they may be supervised or witnessed if desired, by a Government representative.

Custodian:
Air Force - 99

Preparing activity: *
Air Force - 99

Project No. 5865-F018 *

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